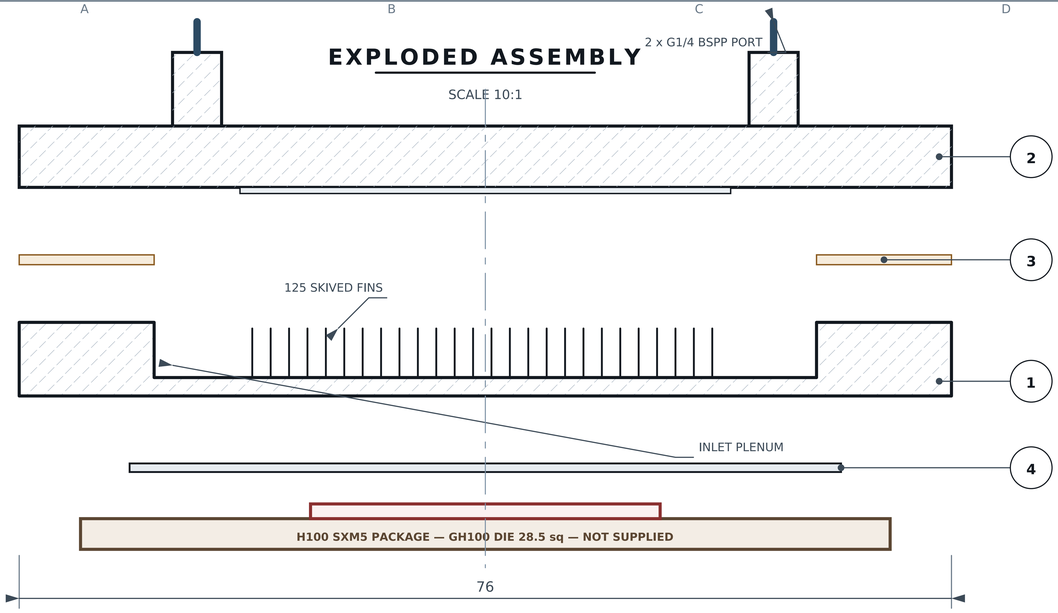
EXPLODED ASSEMBLY

SCALE 10:1



ASSEMBLED HEIGHT 11 mm + 6 PORT BOSS = 17 mm

BILL OF MATERIALS — ASSEMBLY CP-100

ITEM	QTY	PART No.	DESCRIPTION	MATERIAL	NOTE
1	1	CP-101	COLD PLATE BODY, SKIVED FIN	Cu C10200 / C1020	sheet 2
2	1	CP-102	COVER / MANIFOLD, 2 PORTS	Cu C10200 / C1020	sheet 3
3	1	CP-103	BRAZE PREFORM, PERIMETER RING	BCuP-5 (Ag 15%)	0.10 foil
4	1	CP-104	THERMAL INTERFACE, BARE DIE	PTM, k >= 13 W/mK	60 um BLT
5	4	CP-201	SPRING SCREW + BELLEVILLE STACK	A2-70 / 17-7PH	sheet 5
6	2	CP-202	QUICK DISCONNECT, DRIP-FREE	Ni-plated brass	loop side

RESISTANCE BUDGET — PG25, 1.5 L/min

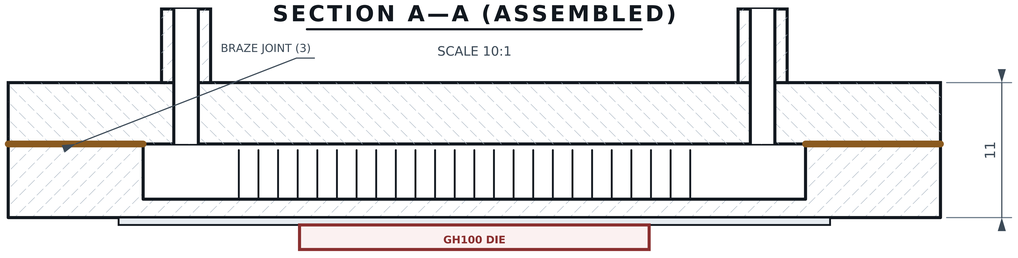
PATH	K/W	dT @ 700 W	SHARE
coolant (convection + sensible)	**0.0141**	9.9 K	48 %
TIM, bare die 60 um k=13	0.0057	4.0 K	19 %
spreading, die to fin field	0.0050	3.5 K	17 %
base 1.5 mm Cu	0.0047	3.3 K	16 %
TOTAL, die to coolant inlet	**0.0295**	**20.7 K**	**100 %**

DIE TEMPERATURE — SAME PLATE, 1.5 L/min

COOLANT IN	H100 700 W	B200 1000 W	GB200 1200 W
32 C	53 C	62 C	67 C
40 C	61 C	70 C	75 C
45 C	**66 C**	**75 C**	**80 C**
50 C	71 C	80 C	85 C

SECTION A—A (ASSEMBLED)

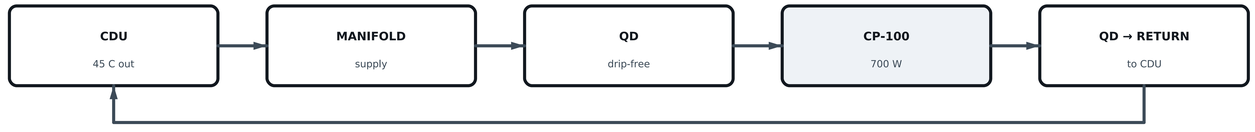
SCALE 10:1



Coolant enters left plenum, crosses 125 channels, leaves right plenum.

LOOP SCHEMATIC

ONE GPU SHOWN



per GPU: 1.5 L/min PG25, 6.1 kPa at the plate, 7.0 K coolant rise

GENERAL NOTES

1. Interpret per ISO 128. Tolerances ISO 2768-mK unless stated. Dimensions in mm.
2. EVERY WETTED PART IS COPPER OR STAINLESS. No aluminium anywhere in the loop — a Cu/Al couple in glycol corrodes the aluminium out within months.
3. Cover is vacuum brazed to the body: no elastomer inside the loop, no serviceable joint. Replace, do not open.
4. SXM5 interface dimensions and mounting preload on sheet 5 are ASSUMED. Confirm against NVIDIA's mechanical specification before manufacture.

INDIA-VC THERMAL HARDWARE	ISO 128 / ISO 2768-mK
TITLE: **D2C COLD PLATE — GENERAL ASSEMBLY** — H100 SXM5 / 700 W / Cu C10200, VACUUM BRAZED / PG25 1.5 L/min	SCALE: 10:1 (THIRD ANGLE PROJECTION) · UNITS: mm
DRAWING No.: **CP-100**	REV: **A** · SHEET: **1 / 6**
DRAWN: AI-ASSISTED DRAFT · CHECKED: — PENDING — · MATERIAL: SEE BOM · FINISH: SEE NOTES · DATE: 2026-09-10	